

ADVANTAGE Section Discussion



Drilling Fluids

Operator Peak Petroleum Indust. Limited	Field Bilabri	Block/Co/Parish OML - 122 / Nigeria	Facility Bilabri DX 1
Well Bilabri DX 1	Rig BULFORD DOLPHIN	Job	Bilabri DX 1

Section Details - 8 1/2" Section - #5

Mud Type Invert Emulsion	Mud System SYN-TEQ (PARA)	Top MD 2770.00 m	Bot MD 3200.00 m
Start-End Date 12/Jan/2006-26/Jan/2006	Open Hole Diam in	Total Days 14	Sect Length 430.00 m
Section Comment Drill 8 1/2" Hole w/SBM To TD 7" Liner			

Section Discussion

Make up 8 1/2" slick BHA and trip in the hole, circulated to condition mud while increase mud weight to 10.9 ppg. In and out. Perform L.O.T at 2620 TVD With 10.9 ppg mud. Maximum Circulating Density shown as 14.82 ppg. Slug pipe & pull out to make new 8 1/2" with Auto-trak BHA. Return to bottom & commenced drilling ahead. From 2782 mts to 3000 mts Problem free. Circulated and condition mud. Raise mud weight to 11.9ppg. Continue to drill. At 3172 mts Circulate and increase mud weight to 12.1 ppg. drill to 3200m.

Observe hi levels gas while connection, Circulate and condition mud. Increase mud weight to 12.5 ppg. Observed electrical power failure. Pull to 9 5/8" shoe. Observed flow into trip tank with pumps off. Shut in Well.

Shut down pressure DP. 160 Psi. Annulus 210 psi. Circulate 10 bbls through line while holding constant 210 psi on choke.

Increase mud weight to 13.5 ppg. Commenced to kill well by weight. (weight method) @ 40 SPM. Circulating pressure 690 psi.

Pumped 90 bbls into drill pipe and lost ability to maintain DP Pressure (640 psi). Observed severe losses. Open choke fully and circulated to degasser at 40 spm. No returns were observed, opened BOP.

Circulated conventionally, observing sever losses. Pump 12.8 ppg mud while mixing 60 LCM mud. Record Total losses close to 450 bbls.

Displace LCM pill to bottom. Without success. Continue to observe losses. Pumped out of the hole to 3111 mts. Circulated. No returns in order to find returns. Continue to pump out of hole to 2884m regained circulation, at 2884 mts. Circulated outgas through choke. Shut in well to mix barite and increase mud weight to 12.9 in surface pits.

Experienced Shut electrical power, restore the same and pump 40 LCM (0) psi annular press. Open well and commenced circulating with 12.9 ppg. Pulled to shoe. Observed gas while static. Attempt to increase mud weight but run out of Barite.

Monitor well on trip tank, observing losses from 5 to 8 bbls/hr. For the next 24 hrs. Till barite reach location.

Stand by filling up well while waiting for SOBM and Barite. Well taking 7 to 8 bbls/hr.

Received mud and increase all surface mud up to 12.9 ppg. Mix LCM pill. Circulated and condition mud to 12.9 ppg. Trip in to 3004m.

Pumped LCM pill and displace the same. Attempt to wash down to 3014m without success. Work out pipe and mix LCM Pill. Pumped the same and displace @ 3004m. Pull to shoe.

Mixed Slug and pulled out to surface and recover Auto-Trak. Make up Slick assembly. and ran in the hole to shoe depth. Mixed SOBM in rig tanks while wait for more mud from boat.

Continue to build up SOBM volume. End up with 700 bbls 12.9 ppg SOBM. While monitoring well through trip tank, mixed 60 bbls LCM pill and displace in order to stop static losses. Trip in from 2705m to 2915m Circulate pill round at 140 gpm. Without losses. Continue to circulated, till gradually establish. Increase pump rate to 360 gpm. Spot extra 60 bbls of LCM pill in pipe while circulate & wash. From 2915 mts to 3057 mts, with full return on shakers. Increase pump rate to 425 gpm. Observed gas level at 25.5 % while circulating at 3057 mts. Continue to wash & ream hole, observed packed off at 3066m Work on pipe, pull free.

Ream and work on pack packed off section. Observed splinters shale caving on shakers. Wiped interval clean. Continue to wash and ream wellbore, hole became progressively more difficult to enter with. Increase volume of caving on shakers. Mix and pumped 60 bbls of LCM & spot on bottom. Pull out from 3105m to 2915m. and became stuck. Jarred free and continue to work on tight spot, 2887m. Face severe losses. Level in raiser dropped. Unable to fill hole by simultaneously filling riser & pumping down drill pipe. Mixed and pumped 2 x 60 bbls of LCM. Filled riser with 11.5 ppg mud to surface. Pull out to 2705 mts inside casing.

Check and monitored well by trip tank. Wellbore, since to be static. 24 hrs losses +1500 bbls.

Continue to pull out of hole. Rig up 7" Liner and run the same. average static losses 15 bbls/hr.

Finished to run 7" liner, displace the same and set in place. Rig up cement line. Pumped spacer plus 35 bbls of cement slurry set shoe at 3061m set packer (top of the liner at 2711m). Slugged pipe and p.o.o.h. Make up 8 1/2" BHA and Trip in for casing test.

While tripping in stand No 72 Air slips set on making Block to bend badly the DP. Lay out bended DP Continue to trip in. Circulate to condition mud weight to 12.9 ppg. Test 7" liner to 4000 psi. Rig down pressure lines and pul to surface and make up 6" slick BHA.